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Department of Electronics & Telecommunication Engineering

Digital VLSI Design

Complete Practice Questions & Detailed Solutions

All 40 Questions | 5–10 Marks Each | With Derivations, Numericals & Diagrams

Q1. NMOS Drain Current Equations

Cut-off Region

Condition: $V_{ps} < V_{tn} \Rightarrow I_{DS} = 0$

Linear (Triode) Region

Condition: $V_{GS} \geq V_{TN}$ and $V_{DS} < (V_{GS} - V_{TN})$

$$\begin{aligned} I_{DS} &= \mu_n \cdot C_{ox} \cdot (W/L) \cdot [(V_{GS} - V_{TN}) \cdot V_{DS} - V_{DS}^2/2] \\ &= K_n \cdot [(V_{GS} - V_{TN}) \cdot V_{DS} - V_{DS}^2/2] \end{aligned}$$

Saturation Region

Condition: $V_{GS} \geq V_{TN}$ and $V_{DS} \geq (V_{GS} - V_{TN})$

$$I_{DS} = (\mu_n \cdot C_{ox} \cdot W) / (2L) \cdot (V_{GS} - V_{TN})^2 = (K_n/2) \cdot (V_{GS} - V_{TN})^2$$

where $K_n = \mu_n \cdot C_{ox} \cdot (W/L)$, V_{TN} = NMOS threshold voltage.

Q2. PMOS Drain Current Equations

Cut-off Region

Condition: $V_{GS} > V_{TP}$ (V_{TP} is negative) $\Rightarrow I_{SD} = 0$

Linear Region

Condition: $V_{GS} \leq V_{TP}$ and $|V_{DS}| < |V_{GS} - V_{TP}|$

$$I_{SD} = \mu_p \cdot C_{ox} \cdot (W/L) \cdot [(V_{SG} - |V_{TP}|) \cdot V_{SD} - V_{SD}^2/2]$$

Saturation Region

Condition: $V_{GS} \leq V_{TP}$ and $V_{SD} \geq (V_{SG} - |V_{TP}|)$

$$I_{SD} = (\mu_p \cdot C_{ox} \cdot W) / (2L) \cdot (V_{SG} - |V_{TP}|)^2$$

Note: V_{TP} is negative (e.g. -0.7 V). Hole mobility $\mu_p \approx \mu_n/2$.

Q3. Short Channel Effects in MOSFETs

- Threshold Voltage Roll-off: V_T decreases as L decreases because source/drain depletion regions share gate charge.
 - Drain Induced Barrier Lowering (DIBL): High V_{DS} lowers the source-channel barrier, reducing V_T and increasing leakage.
 - Hot Carrier Effect (HCE): High lateral field accelerates carriers; they inject into gate oxide causing permanent degradation.
 - Velocity Saturation: Carrier velocity saturates at $\sim 10^7$ cm/s at high fields, limiting I_{DS} .
 - Subthreshold Leakage: Significant current flows even when $V_{GS} < V_T$.
 - Punch-through: Source and drain depletion regions merge; uncontrolled current regardless of gate voltage.
 - Gate Oxide Tunneling: Ultra-thin oxide (< 3 nm) allows direct quantum tunneling.
 - Channel Length Modulation: Effective L decreases with increasing V_{DS} (more pronounced in short channels).
 - Narrow Width Effect: V_T increases as W is reduced due to fringe fields.
 - Surface Scattering: Carriers at Si-SiO₂ interface lose mobility due to scattering.
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Q4. Processes Used in MOSFET Fabrication

1. Oxidation — Thermal growth of SiO₂ (dry or wet)
 2. Photolithography — UV pattern transfer through chrome-on-glass masks
 3. Etching — Wet (chemical) or Dry (RIE/plasma) material removal
 4. Ion Implantation — Doping of source, drain, well, threshold-adjust regions
 5. Diffusion — Thermal spreading of dopants
 6. Chemical Vapor Deposition (CVD) — Poly, SiO₂, Si₃N₄ deposition
 7. Physical Vapor Deposition (PVD) / Sputtering — Metal (Al, W, Cu) deposition
 8. Chemical Mechanical Planarization (CMP) — Surface planarization after metal/dielectric
 9. Epitaxial Growth — Crystalline Si layer on substrate
 10. Annealing — Activates implanted dopants, repairs lattice damage
 11. Silicidation — TiSi₂ or CoSi₂ on gate/S/D for low contact resistance
 12. Cleaning — RCA clean, HF dip to remove native oxide before each step
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Q5. MOSFET Parameters Before and After CVS (Constant Voltage Scaling)

Scaling factor = S ($S > 1$). All dimensions scaled by S ; voltages unchanged.

Parameter	Before CVS	After CVS
Channel Length (L)	L	L / S
Channel Width (W)	W	W / S
Oxide Thickness (t_{ox})	t_{ox}	t_{ox} / S
Supply Voltage (V_{DD})	V_{DD}	V_{DD} (unchanged)
Threshold Voltage (V_t)	V_t	V_t (approx. unchanged)
Doping Concentration (N_a)	N_a	$N_a \times S$
Gate Oxide Cap. (C_{ox})	C_{ox}	$C_{ox} \times S$
Total Gate Capacitance (C)	C	C / S
Drain Current $I_{D_{ssat}}$	I_D	$I_D \times S$
Power per gate (P)	P	$P \times S$
Power Density (P/Area)	P/A	$P \times S^3 / A$ $\uparrow\uparrow$ CRITICAL
Delay ($\tau = CV/I$)	τ	τ / S^2
Area	A	A / S^2

Key problem: Power density rises by S^3 — severe thermal stress. Not thermally safe.

Q6. MOSFET Parameters Before and After CFS (Constant Field Scaling)

All dimensions AND voltages scaled by S . Electric field $E = V/L$ stays constant.

Parameter	Before CFS	After CFS
Channel Length (L)	L	L / S
Channel Width (W)	W	W / S
Oxide Thickness (t_{ox})	t_{ox}	t_{ox} / S
Supply Voltage (V_{DD})	V_{DD}	V_{DD} / S
Threshold Voltage (V_t)	V_t	V_t / S
Doping Concentration (N_a)	N_a	$N_a \times S$
Gate Oxide Cap. (C_{ox})	C_{ox}	$C_{ox} \times S$
Total Gate Cap. (C)	C	C / S
Drain Current $I_{D_{ssat}}$	I_D	I_D / S
Power per gate (P)	P	P / S^2

Power Density (P/Area)	P/A	P/A → CONSTANT ✓
Delay (τ)	τ	τ / S
Area	A	A / S ²

Advantage: Power density is constant — thermally safe. Preferred in practice.

Q7. Masks in NMOS Fabrication (in Order of Use)

1. Active Area Mask (AA / LOCOS Mask): Defines transistor active regions; Si₃N₄ pad protects active areas during field oxidation.
2. Well Mask: Defines n-well or p-well (relevant for CMOS; may be skipped for standalone NMOS on p-substrate).
3. Threshold Adjust Mask: Channel implant (boron/phosphorus) to fine-tune V_T.
4. Gate Mask (Poly Mask): Patterns polysilicon gate; defines channel length.
5. Source/Drain Implant Mask (N+ Mask): Defines n+ source and drain regions via As or P implant.
6. Contact Mask: Opens holes in ILD for electrical contacts to S, G, D.
7. Metal Mask (Metallization Mask): Patterns Al/Cu interconnect layer.

Q8. Numericals — MOSFET Current Equations

Example 1: Linear Region

Given: $K_n = 100 \mu A/V^2$, $V_{TN} = 1 V$, $V_{GS} = 3 V$, $V_{DS} = 1 V$

- $V_{GS} - V_{TN} = 2 V$; $V_{DS} = 1 V < 2 V \Rightarrow$ Linear Region

$$I_{DS} = 100 \times 10^{-6} \times [2 \times 1 - 1^2/2] = 100 \mu A \times 1.5 = 150 \mu A$$

Example 2: Saturation Region

Same device, $V_{DS} = 3 V \Rightarrow V_{DS} > (V_{GS} - V_{TN}) = 2 V \Rightarrow$ Saturation

$$I_{DS} = (K_n/2) \cdot (V_{GS} - V_{TN})^2 = 50 \times 10^{-6} \times 4 = 200 \mu A$$

Example 3: Cut-off

$V_{GS} = 0.5 V < V_{TN} = 1 V \Rightarrow$ Cut-off $\Rightarrow I_{DS} = 0$

Example 4: PMOS Saturation

$K_p = 50 \mu A/V^2$, $V_{TP} = -1 V$, $V_{GS} = -3 V$, $V_{DS} = -4 V$

- $V_{SG} = 3 V$, $V_{SD} = 4 V > (V_{SG} - |V_{TP}|) = 2 V \Rightarrow$ Saturation

$$I_{SD} = (K_p/2) \cdot (V_{SG} - |V_{TP}|)^2 = 25 \times 10^{-6} \times 4 = 100 \mu A$$

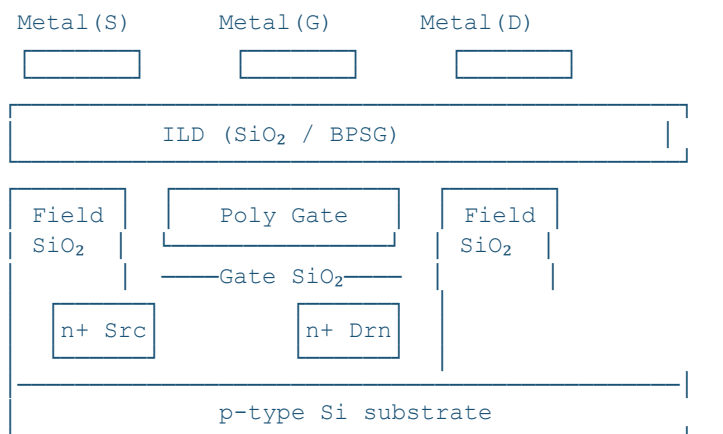
Q9. Numericals — MOSFET Scaling Equations

Original: $L=1\mu\text{m}$, $W=2\mu\text{m}$, $t_{\text{ox}}=10\text{ nm}$, $V_{\text{DD}}=5\text{ V}$, $I_{\text{D}}=1\text{ mA}$, $P=5\text{ mW}$, $\text{Delay}=1\text{ ns}$. Scaling factor $S = 2$.

Parameter	Original	After CVS ($S=2$)	After CFS ($S=2$)
L	$1\mu\text{m}$	$0.5\mu\text{m}$	$0.5\mu\text{m}$
W	$2\mu\text{m}$	$1\mu\text{m}$	$1\mu\text{m}$
t_{ox}	10 nm	5 nm	5 nm
VDD	5 V	5 V (unchanged)	2.5 V
I_{D}	1 mA	2 mA ($\times S$)	0.5 mA ($\div S$)
P	5 mW	10 mW ($\times S$)	1.25 mW ($\div S^2$)
Delay	1 ns	0.25 ns ($\div S^2$)	0.5 ns ($\div S$)
Power Density	—	$\times 8$ ($\times S^3 = \times 8$)	Unchanged

Q10. Cross-Section After Metallization Mask

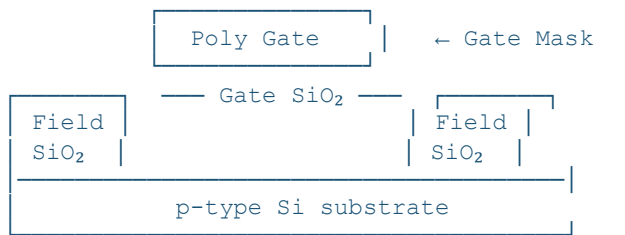
After field oxide, gate, S/D implant, ILD, and contacts, the final metallization step patterns the metal interconnect layer.



Metallization mask (dashed outline) defines which areas of the metal layer are retained. Contacts (W-plugs) connect metal to S/D/Gate.

Q11. Cross-Section After Gate Mask Processing

At this stage: field oxide is grown, gate oxide is grown, polysilicon is deposited and patterned using the Gate Mask. Source/drain implant has NOT yet occurred.



After gate mask: poly is etched everywhere except gate region. Next step = S/D implant (As/P) self-aligned to poly gate.

Q12. Long Channel vs Short Channel MOSFET

Property	Long Channel ($L > 2 \mu\text{m}$)	Short Channel ($L < 1 \mu\text{m}$)
Channel theory	1D gradual channel approx.	2D effects dominate
Threshold V_T	Independent of L and V_{DS}	Depends on L (roll-off) & V_{DS} (DIBL)
I_{DS} vs V_{GS}	Quadratic $\propto (V_{GS} - V_T)^2$	Linear $\propto (V_{GS} - V_T)$ due to vel. sat.
Short-channel effects	Negligible	DIBL, HCE, punch-through, leakage
Subthreshold leakage	Negligible	Significant exponential leakage
Modelling	Simple MOSFET equations	Requires 2D TCAD simulation

Q13. Wet Oxidation vs Dry Oxidation

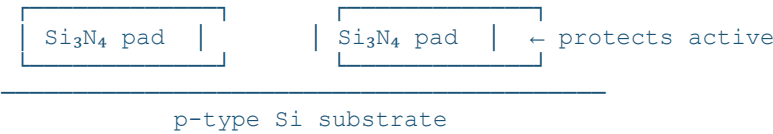
Property	Dry Oxidation	Wet Oxidation
Chemical reaction	$\text{Si} + \text{O}_2 \rightarrow \text{SiO}_2$	$\text{Si} + 2\text{H}_2\text{O} \rightarrow \text{SiO}_2 + 2\text{H}_2$
Growth rate	Slow	Fast (5–10× faster)
Oxide quality	High density, few defects	Lower density, more defects
Si–SiO ₂ interface	Excellent (low Dit)	Good but inferior
Dielectric strength	High	Lower
Temperature	900–1200°C	800–1000°C
Used for	Gate oxide (thin, critical quality)	Field oxide / LOCOS (thick, fast growth)

Gate oxide must be defect-free → dry oxidation. Field oxide needs thickness quickly → wet oxidation.

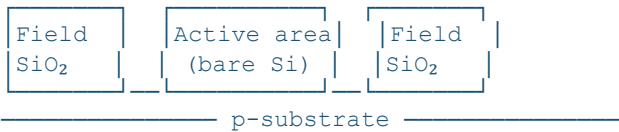
Q14. Cross-Sections of All Masks in NMOS Fabrication

Seven masks in order. Each cross-section shows the cumulative state of the wafer after that mask step.

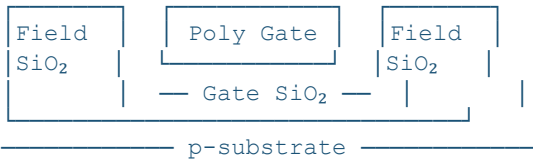
Mask 1: Active Area (LOCOS) Mask



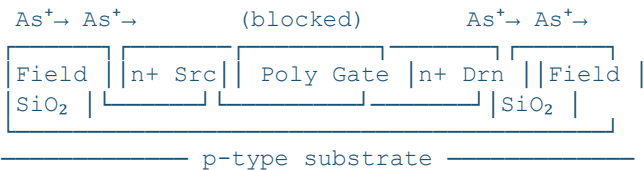
Mask 2: Field Oxidation (after LOCOS)



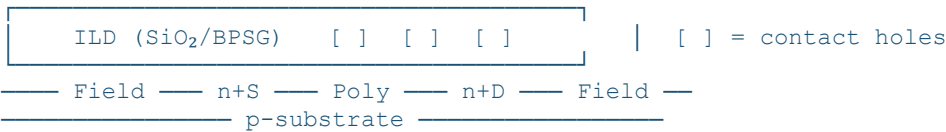
Mask 3: Gate Oxidation + Gate Mask (Poly patterned)



Mask 4: N+ Source/Drain Implant Mask



Mask 5: Contact Mask (holes in ILD)



Mask 6: Metallization Mask (final layer)

See Q10 for detailed cross-section of this stage.

Q15. Hot Carrier Effect (HCE)

In short-channel MOSFETs, the lateral electric field near the drain reaches $\sim 10^6$ V/cm, accelerating channel electrons to very high energies (“hot” carriers).

Mechanism

8. Hot electrons gain energy > 3.1 eV — sufficient to overcome the Si–SiO₂ barrier (3.1 eV).
9. They inject into the gate oxide or create interface traps at the Si–SiO₂ boundary.
10. Trapped charge builds up permanently, shifting device characteristics.

Effects on Device

- V_T shifts over time (increases in NMOS due to trapped negative charge)
- Transconductance g_m degrades
- Drive current I_{DS} reduces over device lifetime

Mitigation

- Lightly Doped Drain (LDD): Spacer oxide + light S/D implant followed by heavy implant reduces peak field
 - Graded junctions: spread field over a wider region
 - Reduce V_{DD} (CFS scaling): lowers lateral field directly
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Q16. Drain Induced Barrier Lowering (DIBL)

In long-channel devices, the gate solely controls the source-channel potential barrier. In short-channel devices, the drain field also affects this barrier.

Mechanism

11. Large V_{DS} causes the drain depletion region to extend towards the source.
12. The electric fringe field from the drain lowers the source–channel potential barrier.
13. This reduces V_T without any change in V_{GS} — the device turns on at a lower V_{GS} .
14. Results in increased subthreshold leakage current at high V_{DS} .

Quantitative Effect

$$\Delta V_T = -\eta \cdot V_{DS}$$

where η is the DIBL coefficient (~ 20 – 100 mV/V depending on device geometry).

Mitigation

Shallow S/D junctions, retrograde well doping, high-k dielectrics, fully-depleted SOI (FD-SOI), FinFET structures.

Q17. Velocity Saturation

In long channels, drift velocity $v = \mu \cdot E$ (linear with field). In short channels, E is very high and velocity saturates at $v_{\text{sat}} \approx 10^7$ cm/s for electrons in Si.

Physics

At high fields, electrons lose kinetic energy to optical phonons faster than they gain it from the field — velocity no longer increases.

Impact on MOSFET Performance

- I_{DSsat} becomes linear in $(V_{\text{GS}} - V_{\text{T}})$ instead of quadratic: $I_{\text{DSsat}} = W \cdot C_{\text{ox}} \cdot v_{\text{sat}} \cdot (V_{\text{GS}} - V_{\text{T}})$
- Saturation voltage $V_{\text{Dsat}} = v_{\text{sat}} \cdot L / \mu$ — lower than classical $(V_{\text{GS}} - V_{\text{T}})$
- Transconductance g_{m} is less dependent on V_{GS}
- Scaling current improvement is reduced compared to long-channel prediction

Q18. VTC of CMOS Inverter — All Critical Points

The Voltage Transfer Characteristic (VTC) plots V_{OUT} vs V_{IN} for the CMOS inverter. It has five distinct regions:

Region	Condition	NMOS State	PMOS State	V_{OUT}
I	$V_{\text{IN}} < V_{\text{TN}}$	Cut-off	Linear	$V_{\text{DD}} = V_{\text{OH}}$
II	$V_{\text{TN}} \leq V_{\text{IN}} < V_{\text{INV}}$	Saturation	Linear	High, decreasing
III	$V_{\text{IN}} \approx V_{\text{INV}}$	Saturation	Saturation	V_{INV} (high gain)
IV	$V_{\text{INV}} < V_{\text{IN}} \leq V_{\text{DD}} + V_{\text{TP}}$	Linear	Saturation	Low, decreasing
V	$V_{\text{IN}} > V_{\text{DD}} + V_{\text{TP}} $	Linear	Cut-off	$0 = V_{\text{OL}}$

Critical Points on VTC

- $V_{\text{OH}} = V_{\text{DD}}$: Maximum output high voltage (Region I)
- $V_{\text{OL}} = 0$ V : Minimum output low voltage (Region V)
- V_{IL} : Input low threshold — slope = -1 on left side. Below V_{IL} , output is valid HIGH.
- V_{IH} : Input high threshold — slope = -1 on right side. Above V_{IH} , output is valid LOW.
- V_{INV} : Inverter switching threshold, where $V_{\text{OUT}} = V_{\text{IN}}$. Peak gain region.
- $\text{NM}_{\text{H}} = V_{\text{OH}} - V_{\text{IH}}$: High noise margin
- $\text{NM}_{\text{L}} = V_{\text{IL}} - V_{\text{OL}}$: Low noise margin

Sketch Description

The VTC starts at (0, V_{DD}), stays flat at V_{DD} until $V_{IN} = V_{TN}$, then drops steeply through the transition region (slope $\gg 1$ at V_{INV}), and reaches 0V when $V_{IN} > V_{DD} + V_{TP}$, staying flat at 0V thereafter. The slope = -1 points mark V_{IL} and V_{IH} .

Q19. Definitions: V_{IL} , V_{IH} , V_{OH} , V_{OL} , V_{INV}

Parameter	Definition	Significance
V_{OH}	Maximum output voltage for logic '1'	= V_{DD} for ideal CMOS
V_{OL}	Minimum output voltage for logic '0'	= 0 V for ideal CMOS
V_{IL}	Max V_{IN} still interpreted as logic '0' ($dV_{OUT}/dV_{IN} = -1$, left side)	Any $V_{IN} \leq V_{IL}$ is a valid '0'
V_{IH}	Min V_{IN} interpreted as logic '1' ($dV_{OUT}/dV_{IN} = -1$, right side)	Any $V_{IN} \geq V_{IH}$ is a valid '1'
V_{INV}	Switching threshold where $V_{OUT} = V_{IN}$	Midpoint of VTC; $\approx V_{DD}/2$ for symmetric design

$$NM_H = V_{OH} - V_{IH}$$

$$NM_L = V_{IL} - V_{OL}$$

Q20. Numerical: V_{OH} , V_{OL} , V_{INV}

Given: $V_{DD} = 3.3$ V, $K_n = 150 \mu A/V^2$, $K_p = 75 \mu A/V^2$, $V_{TN} = 0.7$ V, $V_{TP} = -0.7$ V

V_{OH}

When $V_{IN} < V_{TN}$: NMOS is OFF, PMOS pulls output to V_{DD} .

$$V_{OH} = V_{DD} = 3.3 \text{ V}$$

V_{OL}

When $V_{IN} > V_{DD} + V_{TP} = 3.3 - 0.7 = 2.6$ V: PMOS is OFF, NMOS pulls output to GND.

$$V_{OL} = 0 \text{ V}$$

V_{INV} (both in saturation, $I_{Dn} = I_{Dp}$)

$$\sqrt{K_n} \cdot (V_{INV} - V_{TN}) = \sqrt{K_p} \cdot (V_{DD} - V_{INV} - |V_{TP}|)$$

$$\sqrt{150} \cdot (V_{INV} - 0.7) = \sqrt{75} \cdot (3.3 - V_{INV} - 0.7)$$

$$12.247(V_{INV} - 0.7) = 8.660(2.6 - V_{INV})$$

$$12.247 V_{INV} - 8.573 = 22.516 - 8.660 V_{INV}$$

$$20.907 V_{INV} = 31.089$$

$$V_{INV} \approx 1.487 \text{ V}$$

Q21. CVS vs CFS — Comparison Table

Parameter	CVS (Constant Voltage)	CFS (Constant Field)
Voltage scaling	V unchanged	V scaled by 1/S
ID _{Slin}	×S (increases)	÷S (decreases)
ID _{Ssat}	×S	÷S
Power (P = V·I)	×S (increases)	÷S ² (decreases)
Area	÷S ²	÷S ²
Delay (τ)	÷S ² (better)	÷S
Power Density	×S ³ (SEVERE ↑)	Constant (SAFE ✓)
Oxide electric field	Increases (reliability risk)	Constant (safe)

Q22. NMOS Fabrication Flow with Cross-Sections and Masks

Eight key steps. Cross-sections show cumulative wafer state after each step.

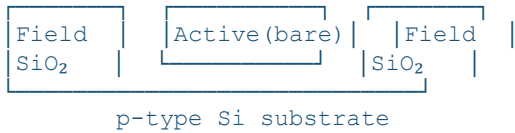
Step 1: Starting Material

P-type Si wafer (boron doped, 10–20 Ω·cm). RCA cleaned.



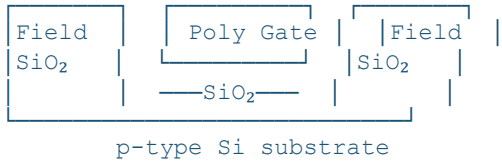
Step 2: Pad Oxide + Si₃N₄ + Active Area Mask → Field Oxidation (LOCOS)

Wet oxidation at 1000°C grows ~500 nm field SiO₂ in unmasked areas.



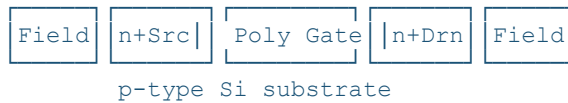
Step 3: Gate Oxidation (dry) + Poly Deposition + Gate Mask

Dry oxidation ~900°C grows 5–10 nm gate SiO₂. LPCVD deposits n+ poly. Gate Mask patterns poly by RIE.



Step 4: N+ Source/Drain Implant (As⁺, self-aligned)

Poly gate acts as self-aligned mask. As⁺ or P⁺ implanted at $\sim 10^{15} \text{ cm}^{-2}$. Annealed to activate.



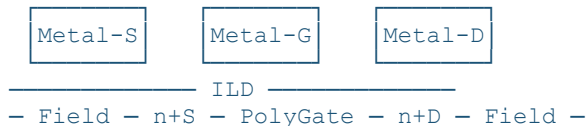
Step 5: ILD (CVD SiO₂) + Contact Mask + W-plug

CVD deposits BPSG/SiO₂. Contact mask opens holes. CMP planarizes. W-plug fills contacts.



Step 6: Metallization (PVD Al/Cu) + Metal Mask

PVD sputters Al. Metal Mask patterns interconnect. Connects S, G, D to external pins.



Q23. Constant Voltage Scaling — Definition & Derivation

Definition: All physical dimensions (L, W, tox, junction depth xj) are scaled by factor S (S > 1), but supply voltage V_{DD} and threshold voltage V_T remain constant.

Derivation of Scaled Parameters

- Scaled Cox: $Cox' = \epsilon_{ox} / (tox/S) = S \cdot Cox$
- Scaled I_{DSsat}: $I' = \mu_n(SCox)(W/S)/[2(L/S)] \cdot (V_{GS}-V_T)^2 = S \cdot I_{DS} \rightarrow \text{current} \times S$
- Gate capacitance: $C' = Cox' \cdot (W/S) \cdot (L/S) = C/S$
- Delay: $\tau' = C'V/I' = (C/S) \cdot V/(SI) = \tau/S^2 \rightarrow \text{delay reduces by } S^2$
- Power: $P' = V \cdot SI = S \cdot P \rightarrow \text{increases by } S$
- Power density: $P'/A' = SP/(A/S^2) = S^3 \cdot P/A \rightarrow \text{increases by } S^3 \text{ (CRITICAL PROBLEM)}$

Impact summary: Faster switching (delay $\div S^2$) but severe thermal crisis (power density $\times S^3$). Not thermally safe for large S.

Q24. Constant Field Scaling — Definition & Derivation

Definition: All dimensions AND voltages are scaled by S to keep the internal electric field $E = V/L$ constant. This ensures oxide reliability and avoids hot carrier effects.

Derivation

- Scaled V_{DD}, V_T: $V' = V/S$

- Scaled Cox: $Cox' = S \cdot Cox$ (same as CVS)
- Scaled I_{DSsat} : $I' = \mu_n(SCox)(W/S)/[2(L/S)] \cdot ((V_{GS}-V_T)/S)^2 = I/S$
- Gate cap: $C' = C/S$
- Delay: $\tau' = (C/S)(V/S)/(I/S) = CV/(SI) = \tau/S \rightarrow$ delay reduces by S
- Power: $P' = (V/S)(I/S) = P/S^2 \rightarrow$ decreases by S^2
- Power density: $P'/A' = (P/S^2)/(A/S^2) = P/A \rightarrow$ CONSTANT
- Oxide field: $E'_{ox} = (V/S)/(tox/S) = V/tox = E_{ox} \rightarrow$ CONSTANT (reliable)

Impact summary: Thermally safe (constant power density), reliable (constant oxide field), delay $\div S$. Preferred for real scaling.

Q25. Proof: Delay Reduces by S^2 (CVS) and S (CFS)

Propagation delay: $\tau_p = (C_L \cdot V_{DD}) / I_{avg}$

CVS Proof ($S > 1$, V unchanged)

- $C'_L = C_L/S$ (capacitance scales as $W \cdot L/tox \rightarrow (1/S^2) \times S = 1/S$)
- $I' = S \cdot I$ (derived in Q23)
- $V' = V$ (unchanged)

$$\tau'_{CVS} = (C_L/S) \cdot V / (S \cdot I) = \frac{C_L \cdot V}{S^2 \cdot I} = \tau / S^2 \quad \checkmark$$

Proved

CFS Proof (V also scales by S)

- $C'_L = C_L/S$
- $I' = I/S$
- $V' = V/S$

$$\tau'_{CFS} = (C_L/S) \cdot (V/S) / (I/S) = \frac{C_L \cdot V}{S \cdot I} = \tau / S \quad \checkmark$$

Proved

Q26. PMOS Transistor Fabrication Flow

PMOS is fabricated on n-type substrate with p+ source and drain. The flow mirrors NMOS with key differences noted.

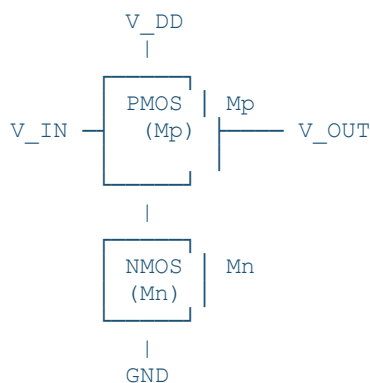
15. Start: n-type Si wafer (phosphorus/arsenic doped)
16. Field Oxidation: Wet ox, ~500 nm SiO₂, LOCOS process (same as NMOS)
17. Gate Oxidation: Dry oxidation, ~5–10 nm gate SiO₂
18. Polysilicon Deposition: LPCVD, p+ doped poly (boron doped) for PMOS gate
19. Gate Mask + Etch: RIE patterns poly gate (same as NMOS)
20. P+ S/D Implant: BF₂⁺ or B⁺ ion implant (~10¹⁵ cm⁻²) self-aligned to gate. Note: Boron for PMOS vs Arsenic for NMOS
21. Anneal: 900°C, activates dopants. BF₂⁺ gives shallow junctions.
22. ILD Deposition (CVD SiO₂) + Contact Mask + W-plug
23. Metallization: PVD Al/Cu + Metal Mask

Key differences from NMOS: n-substrate (not p), B implant (not As/P), V_{TP} is negative, V_{GS} must be negative to turn ON device.

Q27. CMOS Inverter: Circuit Diagram, VTC & Working

Circuit Description

The CMOS inverter consists of one PMOS transistor (M_p) with source at V_{DD} and one NMOS transistor (M_n) with source at GND. Both gates are connected to V_{IN} ; both drains are connected to V_{OUT} .



Working

24. $V_{IN} < V_{TN}$: NMOS OFF ($V_{GS} < V_{TN}$), PMOS ON ($V_{GSp} = V_{IN} - V_{DD} < V_{TP}$). PMOS pulls V_{OUT} to V_{DD} . Output = Logic '1'. No DC path $\rightarrow$ zero static power.
 25. $V_{IN} > V_{DD} + V_{TP}$: PMOS OFF, NMOS ON. NMOS pulls V_{OUT} to GND. Output = Logic '0'. Zero static power.
 26. $V_{IL} < V_{IN} < V_{IH}$: Both transistors in saturation. Transition region. High voltage gain. Brief short-circuit current during switching.
-

Q28. Derivation of V_{IH} and V_{IL}

V_{IL} is where the VTC slope $dV_{OUT}/dV_{IN} = -1$ (left transition: NMOS in saturation, PMOS in linear).

V_{IH} is where the VTC slope = -1 (right transition: PMOS in saturation, NMOS in linear).

At V_{IL} : NMOS in saturation, PMOS in linear. Equate currents and differentiate.

$$I_{Dn} = (K_n/2) (V_{IN} - V_{TN})^2$$

$$I_{Dp} = K_p [(V_{DD} - V_{IN} - |V_{TP}|) (V_{DD} - V_{OUT}) - (V_{DD} - V_{OUT})^2 / 2]$$

Setting $I_{Dn} = I_{Dp}$ and differentiating with respect to V_{IN} (setting $dV_{OUT}/dV_{IN} = -1$):

$$V_{IL} = [3V_{TN} + 2(V_{DD} + V_{TP}) \sqrt{K_n/K_p}] / [3\sqrt{K_n/K_p} + 2]$$

At V_{IH} : PMOS in saturation, NMOS in linear. By symmetry:

$$V_{IH} = V_{DD} - [3|V_{TP}| + 2(V_{DD} - V_{TN})\sqrt{K_p/K_n}] / [3\sqrt{K_p/K_n} + 2]$$

For symmetric inverter ($K_n=K_p$, $|V_{TN}|=|V_{TP}|=V_T$): $V_{IL} \approx V_{DD}/3 + V_T$, $V_{IH} \approx 2V_{DD}/3 - V_T$

Q29. Layout of CMOS Inverter Using Lambda (λ) Rules

Key Lambda Rules (all dimensions in λ)

Layer/Feature	Rule	Value
Poly (gate) width	Minimum width	2λ
Active area width	Minimum width	3λ
Poly over active	Gate extension beyond active	2λ each side
Metal width	Minimum width	3λ
Contact size	Minimum size	$2\lambda \times 2\lambda$
Contact to poly	Minimum spacing	3λ
Metal to metal	Minimum spacing	3λ
Active to active	Minimum spacing	3λ
N-well overlap of p+	Minimum enclosure	3λ

Layout Layer Stack (description)

27. VDD metal rail: Horizontal, 6λ wide, runs at top.
 28. GND metal rail: Horizontal, 6λ wide, runs at bottom.
 29. N-well: Covers upper half of cell (for PMOS); encloses p+ active region by $\geq 3\lambda$.
 30. P+ active (PMOS): In n-well. Width $\geq 3\lambda$. Source contact to VDD via metal.
 31. Vertical poly strip: 2λ wide, runs top-to-bottom crossing both active regions (PMOS and NMOS). This is the gate = input node.
 32. N+ active (NMOS): In p-substrate (lower half). Source contact to GND via metal.
 33. PMOS drain + NMOS drain: Connected by Metal-1 (output node). Connected via contacts ($2\lambda \times 2\lambda$) to drain diffusion.
 34. Input: Poly gate (or metal contact to poly).
 35. Minimum cell width: $\approx 18\lambda$, minimum height: $\approx 24\lambda$.
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Q30. Two-Input NAND Gate and Equivalent Inverter

CMOS NAND: PDN = 2 NMOS in series. PUN = 2 PMOS in parallel.

Reference (W/L) ratio for all transistors in equivalent inverter = 20.

Sizing

- NMOS in PDN (series): 2 NMOS in series doubles resistance. To maintain same drive as reference $\Rightarrow (W/L)_n = 2 \times 20 = 40$
- PMOS in PUN (parallel): 2 PMOS in parallel halves resistance $\Rightarrow$ keep $(W/L)_p = 20$

Equivalent Inverter

- Effective $(W/L)_n = 40/2 = 20$ (two series transistors each $W=40$, effective = $40/2$)
- Effective $(W/L)_p = 20$ (two parallel PMOS, each has $W/L=20$, parallel doesn't increase effective W/L per device)

NAND is preferred over NOR in CMOS because series NMOS (fast electrons) is used for the critical PDN rather than series PMOS (slow holes).

Q31. CMOS Inverter Design for Equal Propagation Delays

Given: $t_{PHL} = t_{PLH} = 20$ ns, $V_{DD} = 5$ V, $K_n = 100 \mu A/V^2$, $K_p = 50 \mu A/V^2$, $V_{TN} = 1$ V, $V_{TP} = -1$ V

Equal Delay Condition

For $t_{PHL} = t_{PLH}$, the pull-down and pull-up drivabilities must be equal:

$$\begin{aligned} K_n \cdot (W/L)_n &= K_p \cdot (W/L)_p \\ 100 \cdot (W/L)_n &= 50 \cdot (W/L)_p \quad \Rightarrow \quad (W/L)_p = 2 \cdot (W/L)_n \end{aligned}$$

Width Calculation (assuming $C_L = 50$ fF, $L = 1 \mu m$)

$$\begin{aligned} W_n &= C_L / [K_n \cdot t_{PHL}] = 50 \times 10^{-15} / [100 \times 10^{-6} \times 20 \times 10^{-9}] \approx 25 \mu m \\ W_p &= 2 \times W_n = 50 \mu m \end{aligned}$$

Result: $(W/L)_n = 25$, $(W/L)_p = 50$ for $1 \mu m$ technology with 50 fF load.

Q32. Two-Input NOR Gate and Equivalent Inverter

CMOS NOR: PDN = 2 NMOS in parallel. PUN = 2 PMOS in series.

Reference $(W/L) = 20$ for equivalent inverter.

Sizing

- NMOS in PDN (parallel): 2 parallel NMOS $\Rightarrow$ keep $(W/L)_n = 20$ each
- PMOS in PUN (series): 2 PMOS in series doubles resistance. To compensate $\Rightarrow (W/L)_p = 2 \times 20 = 40$ each

Equivalent Inverter

- Effective $(W/L)_n = 20$
- Effective $(W/L)_p = 40/2 = 20$

NOR gates are slower than NAND because they place series PMOS (holes, $2\times$ slower) in the critical PUN path.

Q33. Layout of NAND Gate Using Lambda Rules

Layer Description

- 36. VDD rail (top) and GND rail (bottom): Metal-1, 6λ wide.
- 37. N-well (upper half): Contains both PMOS transistors. Each PMOS: p+ active, 3λ wide, shared n-well.
- 38. Two PMOS in parallel: Both gates connected to inputs A and B. Drains connected together by Metal-1 $\rightarrow$ output F.
- 39. Sources of both PMOS connected to VDD rail via contacts.
- 40. Two NMOS in series (lower half): Source of Mn1 to GND; drain of Mn1 = source of Mn2 (shared diffusion, no contact needed to save space); drain of Mn2 = output F.
- 41. Poly strips: Two vertical poly lines crossing NMOS active (gates A and B). Width = 2λ each, spacing $\geq 3\lambda$.
- 42. Output Metal-1: Connects PMOS drains and top NMOS drain to output pin.
- 43. Contacts: $2\lambda \times 2\lambda$, $\geq 1\lambda$ from poly, $\geq 3\lambda$ from active edge.

Total cell size $\approx 28\lambda \times 36\lambda$ for 2-input NAND. NMOS series stack doubles height compared to inverter.

Q34. Layout of NOR Gate Using Lambda Rules

Layer Description

- 44. VDD rail (top) and GND rail (bottom): Metal-1, 6λ wide.
- 45. Two PMOS in series (upper half in n-well): Source of Mp1 at VDD; drain of Mp1 = source of Mp2 (shared p+ diffusion); drain of Mp2 = output F.
- 46. Two poly gates (A, B): Cross PMOS active in series. Each 2λ wide, spacing $\geq 3\lambda$.
- 47. Two NMOS in parallel (lower half): Each NMOS gate connected to A and B. Both drains connected to output F. Both sources to GND.
- 48. NMOS active: Two n+ regions side by side, separated by field oxide. Each has its own poly gate.
- 49. Output connection: Metal-1 connects PMOS series chain drain and both NMOS drains.
- 50. Contacts and lambda rules same as NAND ($2\lambda \times 2\lambda$ contacts, 3λ poly-to-contact, 3λ active-to-active).

Total cell size $\approx 28\lambda \times 36\lambda$. PMOS series stack in the PUN makes NOR wider/taller and slower than NAND.

Q35. Numericals — Noise Margins and Propagation Delay

Numerical 1: Noise Margins

Given: $V_{OH} = 3.3\text{ V}$, $V_{OL} = 0\text{ V}$, $V_{IH} = 2.1\text{ V}$, $V_{IL} = 1.2\text{ V}$

$$NM_H = V_{OH} - V_{IH} = 3.3 - 2.1 = 1.2\text{ V}$$

$$NM_L = V_{IL} - V_{OL} = 1.2 - 0 = 1.2\text{ V}$$

Both margins equal → symmetric design. Noise immunity = ± 1.2 V.

Numerical 2: Propagation Delay t_{PHL}

Given: $C_L = 100$ fF, $V_{DD} = 3.3$ V, $K_n = 150 \mu\text{A/V}^2$, $V_{TN} = 0.7$ V, $(W/L)_n = 10$

$$I_{DSsat} = (K_n/2) \cdot (V_{DD} - V_{TN})^2 = 75 \times 10^{-6} \times (2.6)^2 = 507.5 \mu\text{A}$$

$$t_{\text{PHL}} = C_L \cdot (V_{DD}/2) / I_{\text{avg}} \approx C_L \cdot V_{DD}/2 / (I_{\text{sat}}/2) \\ = 100 \times 10^{-15} \times 1.65 / 253.75 \times 10^{-6} \approx 0.65 \text{ ns}$$

Numerical 3: Propagation Delay t_{PLH}

Given: $K_p = 75 \mu\text{A/V}^2$, $|V_{TP}| = 0.7$ V, same C_L and V_{DD}

$$I_{SDsat} = (K_p/2) \cdot (V_{DD} - |V_{TP}|)^2 = 37.5 \times 10^{-6} \times (2.6)^2 = 253.7 \mu\text{A}$$

$$t_{\text{PLH}} \approx C_L \cdot V_{DD}/2 / (I_{SDsat}/2) = 100 \times 10^{-15} \times 1.65 / 126.8 \times 10^{-6} \approx 1.3 \text{ ns}$$

Propagation Delay Average

$$t_p = (t_{\text{PHL}} + t_{\text{PLH}}) / 2 = (0.65 + 1.3) / 2 \approx 0.975 \text{ ns}$$

Q36. Logic Implementation Styles

Standard CMOS (Complementary)

Uses complementary PDN (NMOS) + PUN (PMOS) network. 2N transistors for N inputs. Full static logic, full rail-to-rail swing, no static power. Most robust and widely used.

Pass Transistor Logic (PTL)

NMOS transistors used as switches. Significantly fewer transistors (e.g. 2-input MUX needs 4 transistors vs 10 in CMOS). Drawback: output degrades by V_{TN} (threshold drop). Cannot cascade without level restoration.

Transmission Gate (TG) Logic

NMOS + PMOS connected in parallel as a bilateral switch. Passes full V_{DD} (PMOS handles high side, NMOS handles low side). Requires complementary control signal (Φ and $\bar{\Phi}$). No threshold drop. Widely used in MUX, XOR implementations.

Pseudo-NMOS

PMOS load always ON (gate tied to GND) as weak pull-up. NMOS PDN implements logic pull-down. Fewer transistors than CMOS. Drawback: static power dissipation when output = '0' (current path from V_{DD} through PMOS to GND via NMOS). Ratioed design required.

Q37. Pass Transistor Logic — Limitations and Solutions

Limitation	Description	Solution
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Threshold voltage drop	NMOS passes $V_{DD}-V_{TN}$, not full V_{DD} . Logic '1' is degraded.	Transmission gates (NMOS+PMOS parallel)
Cascading degradation	Each PTL stage subtracts another V_{TN} . Cascaded stages degrade signal to unusable levels.	Insert CMOS inverter buffer after every 2–3 PTL stages
Short-circuit current	If V_{GS} of next CMOS gate = $V_{DD}-V_{TN}$, PMOS may not fully turn off $\rightarrow$ static I.	Level restorer: weak PMOS feedback transistor
Slow RC delay	Resistance of NMOS switch causes RC delay at output.	Wider transistors; minimize CL with tight layout
No active drive	PTL has no active drive; relies purely on charge sharing.	DCVSL for full-swing differential PTL with active drive

Q38. Layout of a Complex Circuit

Example: Complex gate implementing $F = \overline{AB} + C = AB + \overline{C}$ (using CMOS)

PDN (Pull-Down Network) — NMOS

- Implements $F' = AB + C$ (pull-down when $F = 0$)
- Two NMOS in series (A and B) in parallel with one NMOS (C)
- Series AB: A—drain to B—source, B—drain to output. C: source to GND, drain to output.

PUN (Pull-Up Network) — PMOS — Dual of PDN

- Implements dual: $(A+B)\cdot\overline{C}$
- Two PMOS in parallel (A and B) in series with one PMOS ($\overline{C}$)
- Parallel A,B: Both sources to V_{DD} , both drains connected together. In series with C PMOS to output.

Layout Steps

51. Determine PDN and PUN topology using De Morgan's theorem.
52. Draw N-well for PMOS (upper half), p-substrate for NMOS (lower half).
53. Route poly for all gate inputs (A, B, C) vertically across both NMOS and PMOS active.
54. Connect NMOS series chain with shared diffusion nodes (no contact between series NMOS to save area).
55. Connect PMOS parallel devices with metal-1.
56. Route output metal connecting PMOS drain(s) to NMOS drain(s).
57. Apply all lambda rules: poly 2λ , active 3λ , contacts $2\lambda\times 2\lambda$, metal 3λ , spacings as required.
58. Verify DRC (Design Rule Check) and LVS (Layout vs Schematic).

Q39. MOSFET Regions of Operation — Derivation of Drain Current

Using the Gradual Channel Approximation (GCA): assume the lateral field (x-direction) $\ll$ transverse field (y-direction) so 1D analysis applies along the channel.

Channel Charge Density

At position x along the channel ($V(x)$ = channel potential, 0 at source, V_{DS} at drain):

$$Q_i(x) = -C_{ox} \cdot [V_{GS} - V_T - V(x)]$$

Drift Current

$$I_{DS} = -\mu_n \cdot Q_i(x) \cdot W \cdot (dV/dx) = \mu_n \cdot W \cdot C_{ox} \cdot [V_{GS} - V_T - V(x)] \cdot (dV/dx)$$

Integration (source to drain)

$$I_{DS} \int_0^L dx = \mu_n \cdot W \cdot C_{ox} \int_0^{V_{DS}} (V_{GS} - V_T - V) dV$$

$$I_{DS} = \mu_n \cdot C_{ox} \cdot (W/L) \cdot [(V_{GS} - V_T) \cdot V_{DS} - V_{DS}^2/2] \quad \leftarrow \text{Linear Region}$$

Saturation Onset

Pinch-off occurs when $Q_i(L) = 0 \Rightarrow V_{DS} = V_{GS} - V_T = V_{Dsat}$.

For $V_{DS} > V_{Dsat}$, the channel is pinched off at the drain end; I_{DS} is independent of V_{DS} :

$$I_{DSsat} = \mu_n \cdot C_{ox} \cdot (W/L) / 2 \cdot (V_{GS} - V_T)^2 \quad \leftarrow \text{Saturation Region}$$

Cut-off Region

$V_{GS} < V_T$: No inversion layer formed $\Rightarrow Q_i(x) = 0$ everywhere $\Rightarrow I_{DS} = 0$.

Output Characteristics

Plotting I_{DS} vs V_{DS} for different V_{GS} : curves rise in linear region, then saturate at I_{DSsat} for each V_{GS} . Higher $V_{GS} \Rightarrow$ higher I_{DSsat} . Pinch-off locus: $V_{DS} = V_{GS} - V_T$ (parabola).

Q40. Sources of Power Dissipation in CMOS Inverter

$$P_{total} = P_{dynamic} + P_{short-circuit} + P_{static}$$

1. Dynamic (Switching) Power — Dominant (60–80%)

During 0 $\rightarrow$ 1 transition, C_L charges from V_{DD} . During 1 $\rightarrow$ 0, it discharges to GND. Energy drawn from V_{DD} per full cycle = $C_L \cdot V_{DD}^2$.

$$P_{dynamic} = \alpha \cdot C_L \cdot V_{DD}^2 \cdot f$$

α = activity factor (0 to 1); f = clock frequency.

Reduction strategies: (1) Reduce V_{DD} — quadratic improvement. (2) Reduce C_L with shorter wires, smaller transistors. (3) Reduce f . (4) Clock gating to reduce α .

2. Short-Circuit (Crowbar) Power (5–15%)

During input signal transition ($V_{IL} < V_{IN} < V_{IH}$), both PMOS and NMOS are simultaneously ON, creating a direct V_{DD} -to-GND path.

$$P_{sc} \approx (1/12) \cdot \mu \cdot C_{ox} \cdot (W/L) \cdot (V_{DD} - 2V_T)^3 \cdot (t_r + t_f) / T$$

where t_r , t_f are input rise and fall times, T is clock period.

Reduction: Ensure fast input transitions (small t_r , t_f). Match PMOS and NMOS for equal rise/fall.

3. Static (Leakage) Power — Dominant in Deep Sub-Micron

Sources of leakage even when the circuit is idle (not switching):

- Subthreshold leakage: $I_{sub} = I_0 \cdot \exp((V_{GS} - V_T)/(n \cdot V_T))$ when $V_{GS} < V_T$. Exponentially sensitive to V_T reduction.
- Gate oxide tunneling: Direct quantum tunneling through ultra-thin (<3 nm) gate oxide. Increases exponentially as t_{ox} decreases.
- Junction leakage: Reverse-biased p-n junctions at source/drain have thermally generated leakage current.
- GIDL (Gate-Induced Drain Leakage): High field in gate-drain overlap region causes band-to-band tunneling.

$$P_{static} = I_{leakage} \cdot V_{DD}$$

In modern sub-22 nm nodes, P_{static} can rival or exceed $P_{dynamic}$. Mitigation: multiple V_T devices, body biasing, power gating (sleep transistors), stack effect, multi- V_T design.